Supplementary Information (Online Resource 1)

Article title: How Teams Decide on Human Oversight in AI-Powered Systems: A Mixed-Methods Study

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A comprehensive description of the requirement interrelationships

1) Trustworthiness

One of the central contradictions in the design of AI-powered systems lies in the tension between effective human-AI interaction and the trustworthiness of the system. As AI capabilities advance, there is a growing tendency to increase system autonomy in order to improve performance and reduce operational cost. However, increased automation often comes at the expense of explainability and transparency, all of which are crucial components for gaining trust. When systems operate with limited or no human intervention, users may find it difficult to understand, validate, or challenge the decisions made by the AI. The opacity of complex AI models further exacerbates the issue, as users often struggle to make sense of AI outputs, leading to skepticism and reduced confidence in the system (O'Grady et al., 2022, Autili et al., 2025, Díaz-Rodríguez et al., 2023).

On the other hand, increasing human oversight to enhance trust can reduce the efficiency of the system. Relying too heavily on manual review may introduce delays and inconsistencies, that contradict the very motivations for adopting AI in the first place. This creates a fundamental trade-off: too much automation can undermine trust by removing human judgment and ethical reflection, while too much human control can negate the benefits of automation (Bouderhem, 2024, Amershi et al., 2019, Díaz-Rodríguez et al., 2023).

Trustworthiness cannot be treated as an isolated requirement; it must be understood in relation to how the system interacts with its human users. Achieving an effective balance between automation and human oversight requires explicitly defining human roles and responsibilities, incorporating explainability mechanisms tailored to different user groups, and establishing oversight checkpoints that correspond to the system's critical functions. Integrating these considerations early in the requirements engineering process ensures that trust is maintained rather than compromised by design choices made to optimize other system goals.

2) Ethics – fairness and privacy

Another prominent contradiction emerges between human-AI interaction and fairness. Increasing automation and minimizing human involvement can come at the cost of fairness, especially when decisions are made by opaque AI models trained on biased or unbalanced data. Without sufficient human oversight, these systems risk perpetuating and even amplifying existing inequalities (O'Grady et al., 2022, Bouderhem, 2024, Díaz-Rodríguez et al., 2023). Enhancing fairness requires active human involvement, through manual review, participatory processes, and transparency measures, to detect and mitigate bias (Lepri et al., 2021). Conversely, enhancing fairness through increased human involvement, such as manual review of decisions or bias mitigation interventions, can reduce consistency by introducing human subjectivity, and slow down the system's responsiveness (Bouderhem, 2024, Habiba et al., 2024). This creates a delicate balancing act: minimizing human intervention can threaten fairness, while maximizing it can reduce system efficiency.

Designing systems that support effective human oversight often involves ensuring transparency and explainability, properties that typically require access to sensitive user data or exposure of internal model behavior. However, these practices can conflict directly with privacy-preserving principles, such as data minimization, encryption, and anonymization. In scenarios where strong privacy protections are essential, enforcing transparency may inadvertently compromise user confidentiality or violate legal

requirements such as GDPR. At the same time, limiting transparency to preserve privacy can weaken the ability of humans to audit or contest AI decisions (Bouderhem, 2024, Díaz-Rodríguez et al., 2023, Habiba et al., 2024, Lemke et al., 2024).

Navigating these contradictions requires a thoughtful and context-sensitive approach. Selective transparency, tailored to the roles and responsibilities of different users, can help balance oversight and privacy. Fairness can be enhanced by embedding bias detection and correction mechanisms that operate dynamically, adjusting the level of human intervention based on the sensitivity or risk of the decision context. In some cases, privacy-preserving techniques such as differential privacy or federated learning may allow for both human explainability and user protection (Saifullah et al., 2024).

3) Performance – accuracy, precidtion and efficiency

Human involvement in decision-making, especially in high-stakes domains, necessitates that AI systems provide explainable outputs. This demand for explainability can restrict the use of highly complex models. Many high-performing models, such as deep neural networks, achieve accuracy by operating as opaque "black boxes." Adding layers of explainability or mandating human validation frequently requires simplifying these models, which can lead to loss of predictive power or increased false positives/negatives (Habiba et al., 2024, Lemke et al., 2024). Models designed for high explainability, such as decision trees and rule-based algorithms, allow users to understand how a decision was reached. However, these models tend to be less accurate compared to more complex AI models like deep learning models. Deep learning, while highly effective in pattern recognition and decision-making, is often considered a black-box system, making it difficult for users to interpret or validate the reasoning behind its outputs (Bouderhem, 2024, Habiba et al., 2024, Islam et al., 2022). Deep learning models often rely on complex, non-linear feature interactions that are unintelligible to most users, even experts. While substitute models or post-hoc explanations can be added, they rarely capture the full reasoning of the original model and may create a false sense of transparency. Thus, the desire for human oversight may result in oversimplified explanations that either reduce user trust or distort the true behavior of the system (Habiba et al., 2024, Islam et al., 2022).

From the efficiency standpoint, AI systems are designed to process vast datasets, detect patterns, and make rapid decisions, minimizing latency and maximizing throughput. However, when systems are required to pause for human intervention, review, or approval, this introduces delays, and increased cognitive load (Habiba et al., 2024, Mylrea et al., 2023). Even when oversight is applied selectively, in edge cases, it adds operational overhead and reduces the scalability of the system. Efforts to make AI decisions explainable to humans, through generating interpretable outputs, providing confidence scores, or using post-hoc explanation methods, often require additional processing layers, further increasing computational demands (Habiba et al., 2024, Islam et al., 2022).

4) Safety

At first glance, human supervision appears essential to ensuring safety in AI-powered systems. In many domains, such as healthcare, autonomous mobility, and public infrastructure, human oversight is critical for validating that AI behavior aligns with safety demands. Humans are often better equipped than machines to understand nuance, detect context-specific anomalies, and exercise judgment in ambiguous situations. In this sense, human involvement enhances system safety by acting as a safeguard against erroneous or unsafe autonomous decisions (O'Grady et al., 2022, Autili et al., 2025, Kolarik et al., 2023).

However, in many AI-powered systems, particularly in domains like autonomous driving, critical infrastructure, and defense, safety is frequently associated with fast response time, high precision, and minimal malfunctions (Bouderhem, 2024, Díaz-Rodríguez et al., 2023). Autonomous systems are designed to respond instantly and consistently to potentially dangerous conditions (Mylrea et al., 2023). In contrast, introducing human interaction into the loop, whether for monitoring, approval, or decision-making, can slow response times, introduce uncertainty, or even increase risk due to delayed reactions or inconsistent human judgment (Habiba et al., 2024).

Scenarios that demand real-time decision-making, such as those in cybersecurity, defense, or autonomous systems, highlight the challenge of balancing fast response time with oversight. Autonomous systems are often capable of processing information and responding within milliseconds. However, integrating human oversight in these contexts introduces coordination complexity that can impact responsiveness. The AI Trust Maturity Model, suggested by Mylrea & Robinson emphasizes the importance of carefully calibrating the level of human involvement to ensure timely and effective system behavior without compromising accountability or trust (Mylrea et al., 2023). Furthermore, safety in AI systems often depends on maintaining predictability and operational consistency (Heyn et al., 2022, Gautam et al., 2024).

Resolving these contradictions does not mean eliminating human interaction or compromising safety. Rather, it involves context-aware design choices that align the level of human involvement with the

criticality of the safety requirement. For instance, human validation might be applied retrospectively in non-critical cases, while high-risk decisions remain fully automated but auditable.

5) Explainability & transparency

Explainability and transparency requirements serve as mediating factors that shape the interrelations and trade-offs among multiple system requirements. They mediate the balance between ethical behavior, safety, and performance requirements on one hand, and human oversight on the other, as discussed in the preceding sections. However, different users require different kinds of explanations: developers seek technical diagnostics, while end-users often require high-level, intuitive narratives. Designing one-size-fits-all explainability interfaces for human-AI interaction proves challenging, and in many cases, increased human involvement demands explanations that are not natively supported by the underlying AI, leading to compromises in fidelity or usability (Habiba et al., 2025, Islam et al., 2022, Lemke et al., 2024).

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